RAG Learning

C4 — Predicting and Identifying Reactions and Products

OCR GCSE Chemistry A (J248) — Gateway Science · Papers 2 & 4

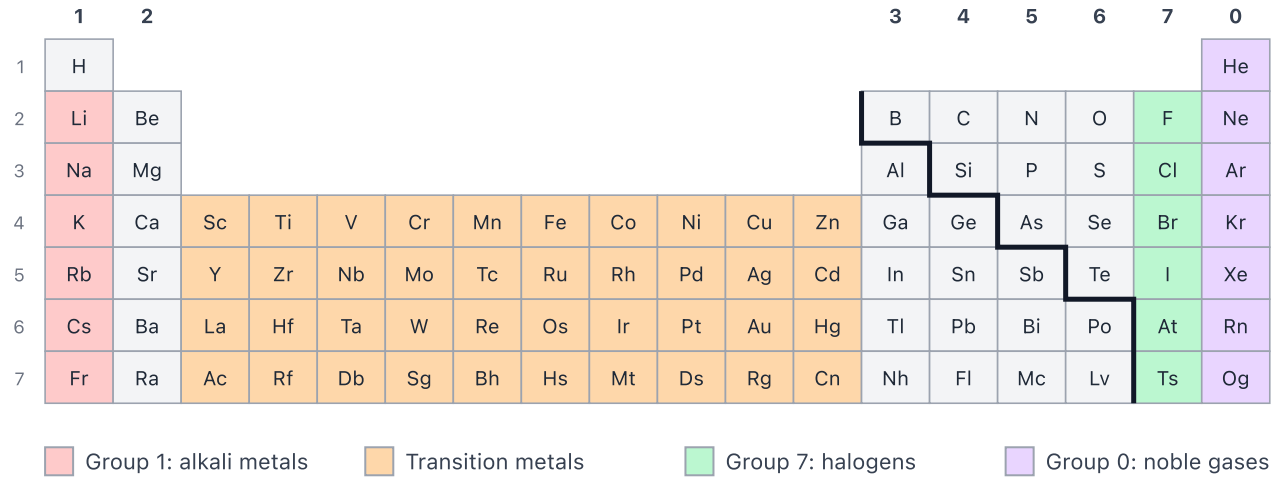
Topic C4 is assessed on **Paper 2** (Foundation) and **Paper 4** (Higher); those papers also assume everything in C1–C3. None of the C4 learning outcomes is Higher-tier only, so everything here is for both tiers — but where an **ionic equation** or **half equation** appears, writing it is a Higher skill from C3 and is marked **(HT)**.

C4.1 Predicting Chemical Reactions

> In the **Periodic Table**, elements are arranged in order of **atomic (proton) number**. Elements in the same **group** (vertical column) have the **same number of outer-shell electrons**, so they have **similar chemical properties**. The **period** (horizontal row) tells you the number of occupied electron shells.

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The Periodic Table: groups and blocks you must know



Thick stepped line: metals to the left, non-metals to the right. Period number = number of occupied shells.

Because reactions involve the outer electrons, you can **predict** how an unfamiliar element will behave from its group, and how strongly it will react from its position in the group (C4.1d).

Group 1 — the alkali metals (C4.1a, C4.1b)

Physical properties

- Soft — can be cut with a knife; shiny when freshly cut, quickly tarnish in air
- **Low densities** (Li, Na, K float on water)
- **Low melting points** for metals, **decreasing** down the group (Li 181 °C, Na 98 °C, K 63 °C)
- Good conductors of heat and electricity

Chemical properties

- React vigorously with **water** → metal hydroxide (alkaline solution) + hydrogen
- React with **chlorine** → white metal chloride (ionic salt)
- React with **oxygen** → metal oxide
- Form **1+ ions** (Li^+, Na^+, K^+) in compounds, which are white and dissolve to give colourless solutions
- Stored under oil to keep air and water out

$$2Na(s) + 2H_2O(l) \rightarrow 2NaOH(aq) + H_2(g)$$

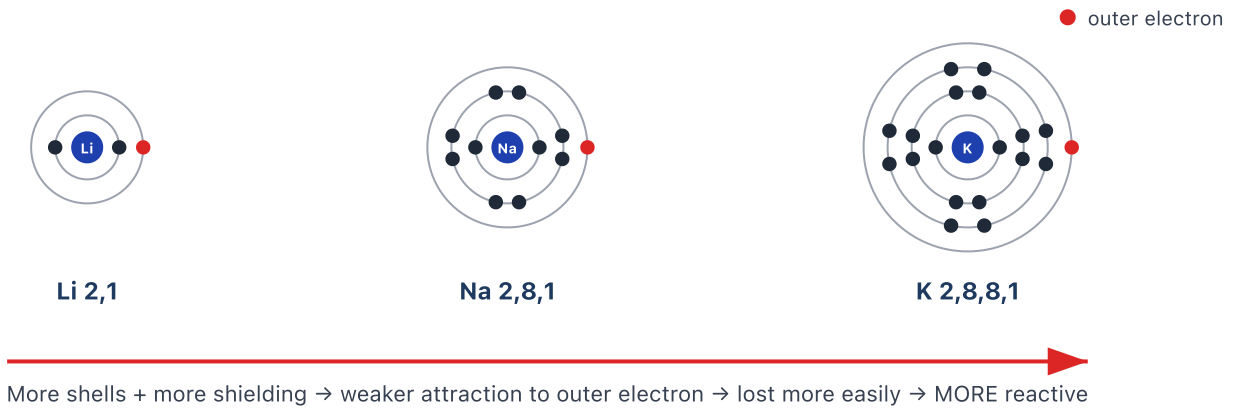
$$2K(s) + Cl_2(g) \rightarrow 2KCl(s) \qquad 4Li(s) + O_2(g) \rightarrow 2Li_2O(s)$$

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Metal	Observations with cold water
Lithium	Floats, fizzes steadily, slowly gets smaller
Sodium	Floats, melts into a silvery ball, fizzes rapidly, whizzes around the surface
Potassium	Floats, melts, fizzes violently; the hydrogen ignites with a **lilac flame**; may spit

Universal indicator added to the water turns **purple/blue** because the hydroxide formed is alkaline.

Group 1: one outer electron, further from the nucleus down the group



Explaining the trend. All Group 1 atoms have **one outer electron**, which they lose to form a 1+ ion with a full outer shell. Going down the group the atoms have **more shells**, so the outer electron is **further from the nucleus** and **more shielded** by inner electrons. The attraction between the nucleus and the outer electron is **weaker**, so the electron is **lost more easily** and the metal is **more reactive**.

> **Worked Example:** Predict how rubidium (below potassium) reacts with water.
> It reacts **even more violently** than potassium (possibly explosively), forming **rubidium hydroxide** and **hydrogen**: $2Rb + 2H_2O \rightarrow 2RbOH + H_2$. Its melting point will be lower than potassium's (below 63 °C) and it forms Rb^+ ions.

Group 7 — the halogens (C4.1a, C4.1b)

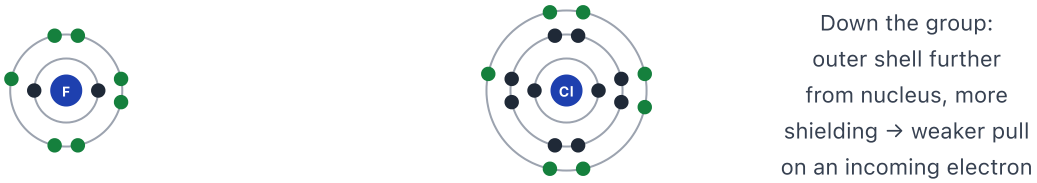
Halogen	Formula	State at room temperature	Colour

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Fluorine	F_2	Gas	Pale yellow
Chlorine	Cl_2	Gas	Pale green (yellow-green)
Bromine	Br_2	Liquid	Red-brown liquid, orange vapour
Iodine	I_2	Solid	Grey-black solid, purple vapour

- Non-metals made of **diatomic molecules** (X_2); toxic and corrosive.
- **Melting and boiling points increase** down the group (larger molecules → stronger intermolecular forces), and the colour gets **darker**.
- React with metals to form **metal halides** (salts containing 1– halide ions): $2Na + Cl_2 \rightarrow 2NaCl$; $2Fe + 3Cl_2 \rightarrow 2FeCl_3$.
- React with hydrogen to form **hydrogen halides** (e.g. HCl), which dissolve in water to form acids.
- **Reactivity decreases** down the group.

Group 7: seven outer electrons, need to GAIN one



F 2,7

Cl 2,8,7

Each atom gains one electron to form a 1– halide ion (F^-, Cl^-) with a full outer shell.

Explaining the trend. Halogen atoms have **seven outer electrons** and react by **gaining one electron** to form a 1– ion. Down the group the outer shell is **further from the nucleus** and more shielded, so the nucleus attracts an incoming electron **less strongly**. An electron is **gained less easily**, so reactivity **decreases**.

> **Exam Tip:** Group 1 and Group 7 trends go in **opposite** directions, but the explanation is the same idea: distance and shielding weaken the attraction between nucleus and outer

electrons. Group 1 → *losing* gets easier (more reactive); Group 7 → *gaining* gets harder (less reactive). Always say "outer electron" and "nucleus", not just "the atom is bigger".

Halogen displacement reactions (PAG C1)

A **more reactive halogen displaces a less reactive halogen** from a solution of its salt (a halide). This is the OCR example activity for **PAG C1 — Reactivity trend**: add halogen solutions (chlorine water, bromine water, iodine solution) to solutions of potassium chloride, bromide and iodide and look for a colour change.

Halogen added ↓ / Salt solution →	KCl (Cl^-)	KBr (Br^-)	KI (I^-)
Chlorine water	—	✓ turns **orange** (Br_2 formed)	✓ turns **brown** (I_2 formed)
Bromine water	× no reaction	—	✓ turns **brown** (I_2 formed)
Iodine solution	× no reaction	× no reaction	—

$$Cl_2(aq) + 2KBr(aq) \rightarrow 2KCl(aq) + Br_2(aq)$$

$$Br_2(aq) + 2KI(aq) \rightarrow 2KBr(aq) + I_2(aq)$$

(HT) ionic: $Cl_2 + 2Br^- \rightarrow 2Cl^- + Br_2$ (chlorine is reduced; bromide ions are oxidised)

Worked Example: Predict whether astatine (below iodine) would react with potassium iodide solution.
Astatine is **less reactive** than iodine (reactivity decreases down Group 7), so it **cannot displace** iodine — **no reaction**. We would also predict astatine is a dark solid with a higher melting point than iodine.

Group 0 — the noble gases (C4.1a, C4.1b)

- Colourless gases at room temperature, made of **single atoms** (monatomic): He, Ne, Ar, Kr, Xe, Rn.
- **Very unreactive (inert)** because they have a **full outer shell** (helium 2, the others 8). They do not need to lose, gain or share electrons, so they rarely form compounds or molecules.
- **Boiling points increase** down the group (heavier atoms, stronger forces between them); density also increases.
- Uses that depend on unreactivity/low density: helium in balloons and airships; argon in filament lamps and as a shield gas in welding; neon in advertising signs.

Noble gas	Electron arrangement	Boiling point (°C)
Helium	2	−269
Neon	2,8	−246
Argon	2,8,8	−186
Krypton	(full outer shell of 8)	−153

Transition metals (C4.1c)

The transition metals are the block of metals in the middle of the Periodic Table, between Groups 2 and 3 (e.g. iron, copper, nickel, manganese, chromium). Compared with Group 1 metals they have:

Property	Transition metals (e.g. iron)	Group 1 metals (e.g. sodium)
Melting point	**High** (Fe 1538 °C) — except mercury	Low (Na 98 °C)
Density	**High** (Fe 7.9 g/cm³)	Low (Na 0.97 g/cm³)
Strength / hardness	Strong and hard	Soft
Reactivity	**Much less reactive**: react slowly (or not at all) with water and oxygen	Very reactive

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Ions	Form **ions with different charges**, e.g. Fe^{2+} and Fe^{3+}; Cu^{+} and Cu^{2+}	Only 1+ ions
Compounds	**Coloured** compounds and solutions	White compounds, colourless solutions
Catalysts	Many are **useful catalysts**	Not used as catalysts

Coloured ion / compound	Colour	Catalyst	Used for
Cu^{2+} (e.g. copper(II) sulfate)	Blue	Iron, Fe	Haber process (making ammonia)
Fe^{2+} (e.g. iron(II) sulfate)	Pale green	Nickel, Ni	Hydrogenation of vegetable oils (alkenes)
Fe^{3+} (e.g. iron(III) chloride)	Yellow / orange-brown	Vanadium(V) oxide, V_2O_5	Contact process (making sulfuric acid)
MnO_4^- (potassium manganate(VII))	Purple	Manganese(IV) oxide, MnO_2	Decomposing hydrogen peroxide: $2H_2O_2 \rightarrow 2H_2O + O_2$

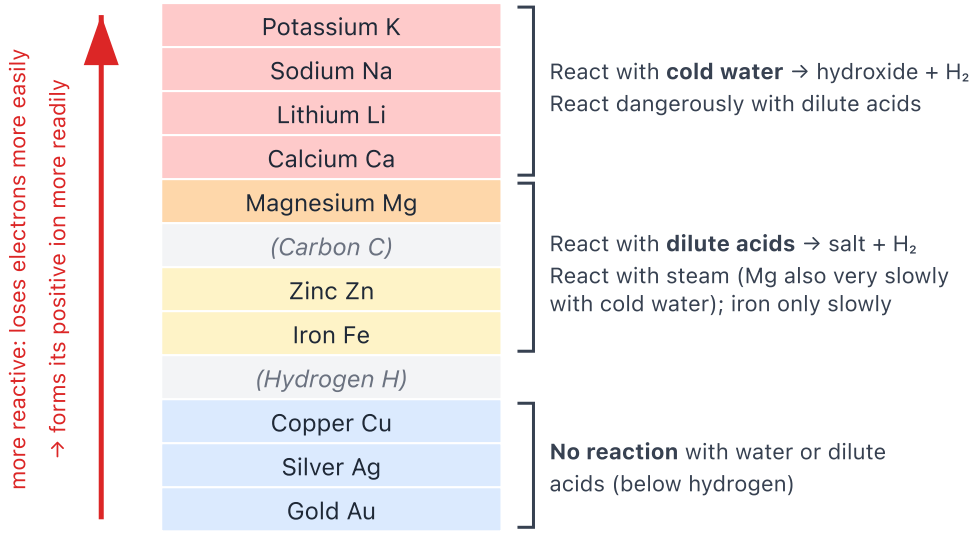
PAG link — investigation of transition metals (PAG C1, C5, C8): e.g. compare the colours of transition metal salt solutions, test them with sodium hydroxide (coloured precipitates, see C4.2), or measure how fast oxygen is given off from hydrogen peroxide with different transition metal oxides as catalysts.

Reactivity of metals with water and dilute acids (C4.1e)

> When a metal reacts it **loses electrons to form a positive ion**. The **more easily a metal forms its positive ion**, the **more reactive** it is. The **reactivity series** lists metals in order of this tendency.

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The reactivity series of metals



Metal	With cold water	With dilute hydrochloric acid
Potassium, sodium, lithium	Vigorous fizzing (see Group 1)	Violent — never done in school
Calcium	Fizzes steadily; solution goes cloudy ($Ca(OH)_2$ is only slightly soluble)	Very vigorous fizzing
Magnesium	A few bubbles very slowly	Rapid fizzing, gets hot, ribbon disappears
Zinc	No visible reaction	Steady bubbling
Iron	No visible reaction (rusts slowly over days)	Slow bubbling
Copper	No reaction	No reaction

$$Ca(s) + 2H_2O(l) \rightarrow Ca(OH)_2(aq) + H_2(g) \qquad Mg(s) + 2HCl(aq) \rightarrow MgCl_2(aq) + H_2(g)$$

(HT) $Mg \rightarrow Mg^{2+} + 2e^-$ The metal atom is oxidised to its positive ion; H^+ ions are reduced: $2H^+ + 2e^- \rightarrow H_2$

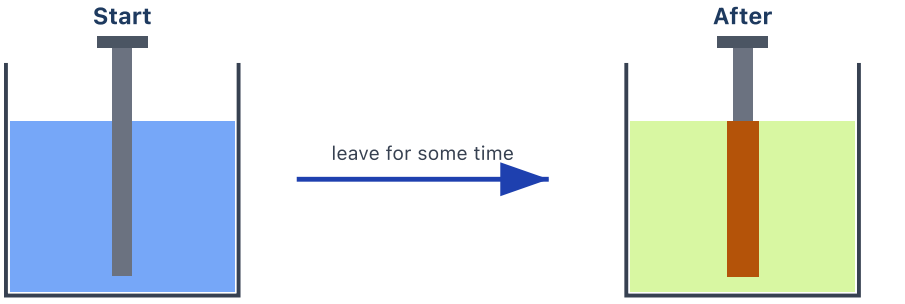
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PAG link (PAG C1, C7, C8): "Reaction of metals with water, dilute hydrochloric acid". Compare the rate of bubbling, measure the temperature rise, or collect the hydrogen in a gas syringe over a fixed time. Keep it a fair test: same mass and surface area (e.g. powder or same-length strips), same acid volume and concentration, same starting temperature.

Deducing an order of reactivity from results (C4.1f)

A **more reactive metal displaces a less reactive metal** from a solution of its compound. The more reactive metal forms ions and goes into solution; the less reactive metal comes out as the solid element.

Displacement: iron nail in copper(II) sulfate solution



$Fe(s) + CuSO_4(aq) \rightarrow FeSO_4(aq) + Cu(s)$

Metal added ↓ / Solution →	Magnesium sulfate	Zinc sulfate	Iron(II) sulfate	Copper(II) sulfate
Magnesium	—	✓	✓	✓
Zinc	×	—	✓	✓
Iron	×	×	—	✓
Copper	×	×	×	—

Count the ticks: the metal that displaces the most others is the most reactive. Order: **Mg > Zn > Fe > Cu**. A bigger **temperature rise** in a displacement reaction also shows a bigger difference in reactivity.

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Worked Example: Three metals X, Y and Z were added to dilute acid. X: rapid bubbling, temperature rose 12 °C. Y: no bubbles. Z: slow bubbling, temperature rose 3 °C. Then Z was placed in a solution of Y's sulfate and became coated. Deduce the order.
X reacts fastest → most reactive. Z reacts slowly with acid and displaces Y → Z is more reactive than Y. Y does not react with acid → below hydrogen.
Order: **X > Z > Y**. (X could be magnesium, Z iron, Y copper.)

PAG link (PAG C1, C7, C8): "Displacement reactions involving metals and metal salts" — on a spotting tile or in test tubes, or in a polystyrene cup measuring the temperature change.

Predicting reactions from the Periodic Table (C4.1d)

- **Metals** (left and centre) tend to **lose electrons** and form positive ions; their oxides are **basic**. **Non-metals** (right) tend to **gain or share electrons**; their oxides are usually **acidic**.
- Elements in the **same group** react in the **same way** and form compounds with similar formulae (NaCl, KCl, RbCl; MgO, CaO).
- Reactivity of metals **increases down** a group and decreases across a period (Na > Mg > Al). Reactivity of non-metals in Group 7 **decreases down** the group.
- A metal and a non-metal far apart in the table (e.g. K and F) react most vigorously.

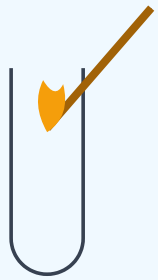
Worked Example: Caesium is in Group 1 and bromine in Group 7. Predict the formula of the product and the type of reaction.
Cs forms Cs^+; Br forms Br^- → caesium bromide, **CsBr**, an ionic white solid. As caesium is very low in Group 1 it reacts extremely vigorously: $2Cs + Br_2 \rightarrow 2CsBr$.

C4.2 Identifying the Products of Chemical Reactions

Tests for gases (C4.2a)

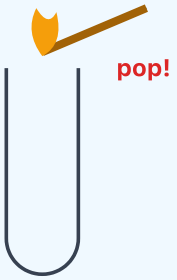
Tests for four gases

Oxygen, O_2



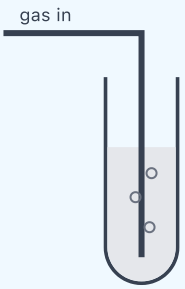
Put in a glowing splint: **it RELIGHTS**

Hydrogen, H_2



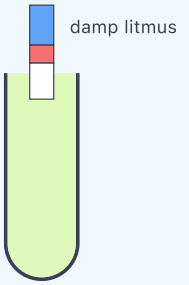
Hold a lighted splint at the mouth: **SQUEAKY POP**

Carbon dioxide, CO_2



Bubble through limewater: **turns MILKY (cloudy)**

Chlorine, Cl_2



Damp blue litmus paper turns red, then **BLEACHED white**

Gas	Test	Positive result	Explanation
Oxygen	Glowing splint into the gas	Splint **relights**	Oxygen supports combustion
Hydrogen	Lighted splint at the mouth of the tube	**Squeaky pop**	Hydrogen burns rapidly: $2H_2 + O_2 \rightarrow 2H_2O$
Carbon dioxide	Bubble gas through limewater (calcium hydroxide solution)	Limewater turns **milky/cloudy**	Insoluble calcium carbonate forms: $Ca(OH)_2 + CO_2 \rightarrow CaCO_3 + H_2O$
Chlorine	Hold damp blue litmus paper in the gas (in a fume cupboard)	Turns red then **bleached white**	Chlorine is acidic and a bleach; paper must be damp

Exam Tip: Don't mix up the splints: **glowing** splint for oxygen (relights), **lighted** (burning) splint for hydrogen (squeaky pop). Say "milky" or "cloudy" for limewater, never "clear" (that is what you start with).

Tests for metal ions using sodium hydroxide (C4.2b)

Add a few drops of dilute sodium hydroxide solution to a solution of the unknown compound. Many metal hydroxides are insoluble and form a coloured **precipitate**. Then add **excess** NaOH to tell

calcium and zinc apart.

Adding a few drops of sodium hydroxide solution to metal ions



Ca^{2+}
white ppt
stays in excess

Zn^{2+}
white ppt
dissolves in excess

Cu^{2+}
blue ppt

Fe^{2+}
green ppt
(browns in air)

Fe^{3+}
orange-brown ppt

ppt = precipitate (an insoluble solid forming in a solution). Add excess NaOH to tell Ca^{2+} from Zn^{2+}.

Metal ion	Few drops of NaOH(aq)	Excess NaOH(aq)	(HT) Ionic equation
Calcium, Ca^{2+}	White precipitate	Precipitate remains	$Ca^{2+} + 2OH^- \rightarrow Ca(OH)_2$
Zinc, Zn^{2+}	White precipitate	**Dissolves** → colourless solution	$Zn^{2+} + 2OH^- \rightarrow Zn(OH)_2$
Copper(II), Cu^{2+}	**Blue** precipitate	Remains	$Cu^{2+} + 2OH^- \rightarrow Cu(OH)_2$
Iron(II), Fe^{2+}	**Green** precipitate (turns orange-brown at the surface on standing in air)	Remains	$Fe^{2+} + 2OH^- \rightarrow Fe(OH)_2$
Iron(III), Fe^{3+}	**Orange-brown** (rust-coloured) precipitate	Remains	$Fe^{3+} + 3OH^- \rightarrow Fe(OH)_3$

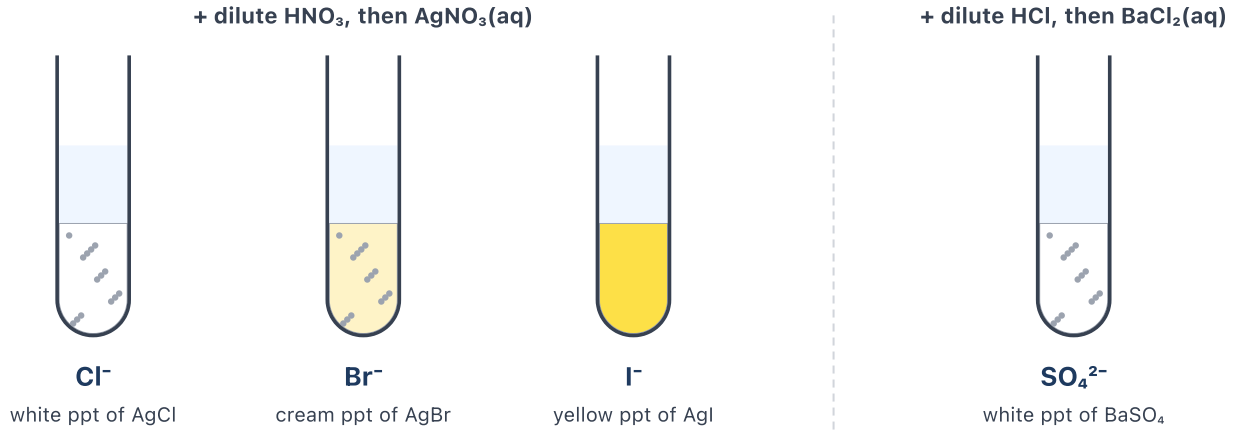
Tests for negative ions (C4.2b)

Anion	Test	Positive result

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Carbonate, CO_3^{2-}	Add dilute acid (e.g. HCl); bubble any gas through limewater	**Fizzing**; gas (CO_2) turns limewater **milky**
Chloride, Cl^-	Add dilute **nitric acid**, then **silver nitrate** solution	**White** precipitate (AgCl)
Bromide, Br^-		**Cream** precipitate (AgBr)
Iodide, I^-		**Yellow** precipitate (AgI)
Sulfate, SO_4^{2-}	Add dilute **hydrochloric acid**, then **barium chloride** solution (or dilute nitric acid then barium nitrate)	**White** precipitate ($BaSO_4$)

Precipitate tests for halide and sulfate ions



The acid is added first to remove carbonate ions, which would also give a precipitate.

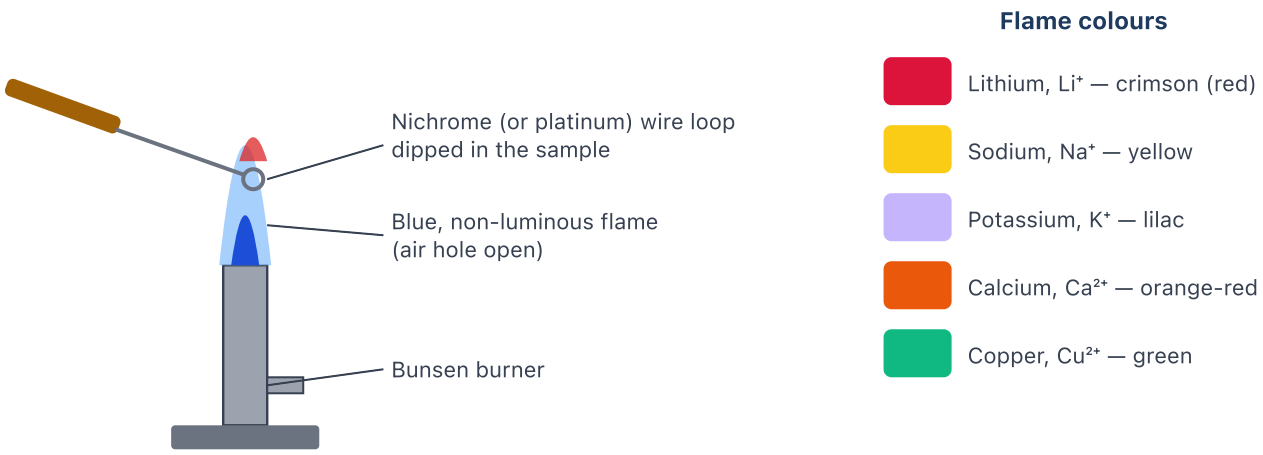
(HT) $Ag^+(aq) + Cl^-(aq) \rightarrow AgCl(s)$ $Ba^{2+}(aq) + SO_4^{2-}(aq) \rightarrow BaSO_4(s)$ $2H^+(aq) + CO_3^{2-}(aq) \rightarrow H_2O(l) + CO_2(g)$

Exam Tip: Why acidify? Carbonate ions would also form a precipitate with Ag^+ or Ba^{2+} (a false positive), so the acid removes them as CO_2 first. Use **nitric** acid for the silver nitrate test (HCl would add Cl^- ions) and **hydrochloric** acid for the barium chloride test (H_2SO_4 would add SO_4^{2-} ions).

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Flame tests (C4.2c, C4.2e)

Flame test: method and colours



Clean the loop first in concentrated HCl and hold it in the flame until it gives no colour.

1. **Clean** a nichrome (or platinum) wire loop by dipping it in concentrated hydrochloric acid and holding it in a hot, blue Bunsen flame until it gives **no colour**.
2. Dip the clean loop in the acid and then into the solid sample (or the solution).
3. Hold the loop in the **edge of a blue (non-luminous) flame** — a yellow flame would mask the colour.
4. Record the colour; clean the loop again before the next sample.

Metal ion	Flame colour
Lithium, Li^+	Crimson (red)
Sodium, Na^+	Yellow (orange-yellow)
Potassium, K^+	Lilac
Calcium, Ca^{2+}	Orange-red (brick red)
Copper, Cu^{2+}	Green (blue-green)

Exam Tip: Flame tests only work well for a **single** metal ion. In a mixture, one colour (especially sodium's intense yellow) can **mask** the others. Lithium's crimson and calcium's

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orange-red are easy to confuse — instrumental methods solve both problems.

Identifying species from test results (C4.2d) — PAG C5

PAG C5 — Identification of species: identify an unknown compound using cation tests, anion tests and flame tests. Test separate small portions of the solution so that one test does not interfere with another, and test for carbonate before adding acid for other anion tests.

Worked Example: A white solid dissolves in water. Flame test: lilac. Add dilute nitric acid then silver nitrate: yellow precipitate. Identify it.
Lilac → K^+. Yellow precipitate with acidified $AgNO_3$ → I^-. The compound is **potassium iodide, KI**.

Worked Example: A green solution gives a green precipitate with sodium hydroxide. Another portion, acidified with HCl and treated with barium chloride, gives a white precipitate. Identify it.
Green precipitate → Fe^{2+}. White precipitate with acidified $BaCl_2$ → SO_4^{2-}. The compound is **iron(II) sulfate, $FeSO_4$**.

Worked Example: Solid Q fizzes with dilute acid, giving a gas that turns limewater milky. A solution of Q in nitric acid gives a white precipitate with NaOH that dissolves in excess. Identify Q.
Fizzing + CO_2 → CO_3^{2-}. White precipitate soluble in excess NaOH → Zn^{2+}. Q is **zinc carbonate, $ZnCO_3$**.

Instrumental methods of analysis (C4.2f)

Chemical tests use reagents and observations by eye. **Instrumental methods** use machines, e.g. **flame emission spectroscopy** (identifies metal ions from the exact wavelengths of light they emit, and measures their concentration) and **mass spectrometry** (measures relative molecular mass and helps identify compounds).

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Advantage	Meaning
Sensitive	Detect and identify **very small amounts** or very low concentrations of a substance; only a tiny sample is needed
Accurate	Give reliable, objective identification and precise quantitative values (no judging colours by eye); can analyse mixtures
Rapid (speed)	Results in seconds or minutes; can be automated to run many samples

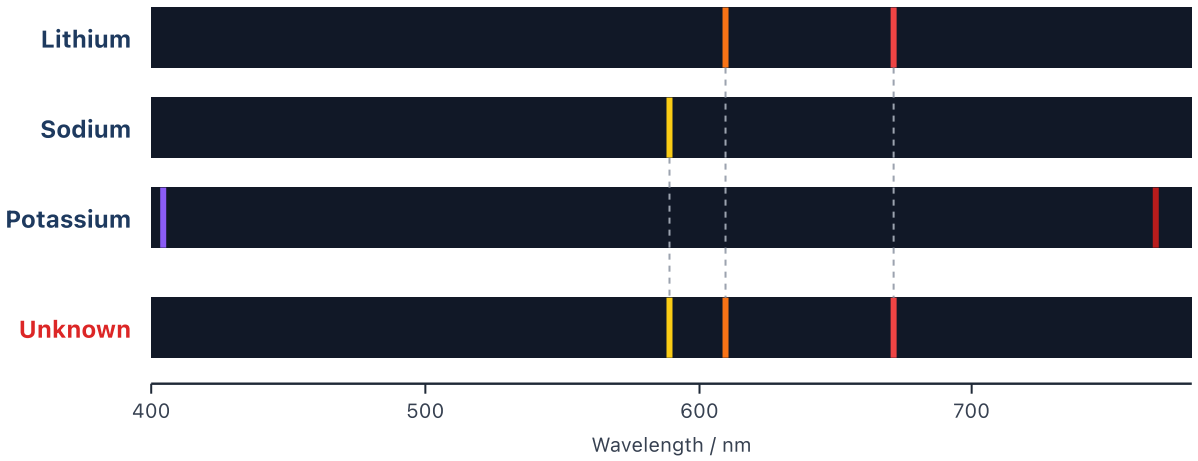
Disadvantages: the instruments are **expensive**, need trained operators, and results must be compared with **reference data** from known samples.

Interpreting instrumental results (C4.2g)

An instrumental result is interpreted by **comparing it with a reference set of data** presented in the same form (chart or table). If the pattern matches, the substance is identified.

Flame emission spectra. Each metal ion gives its own pattern of coloured lines (a "fingerprint"). Lines in an unknown spectrum that match a reference show which ions are present — even in a mixture.

Matching an unknown emission spectrum to reference spectra

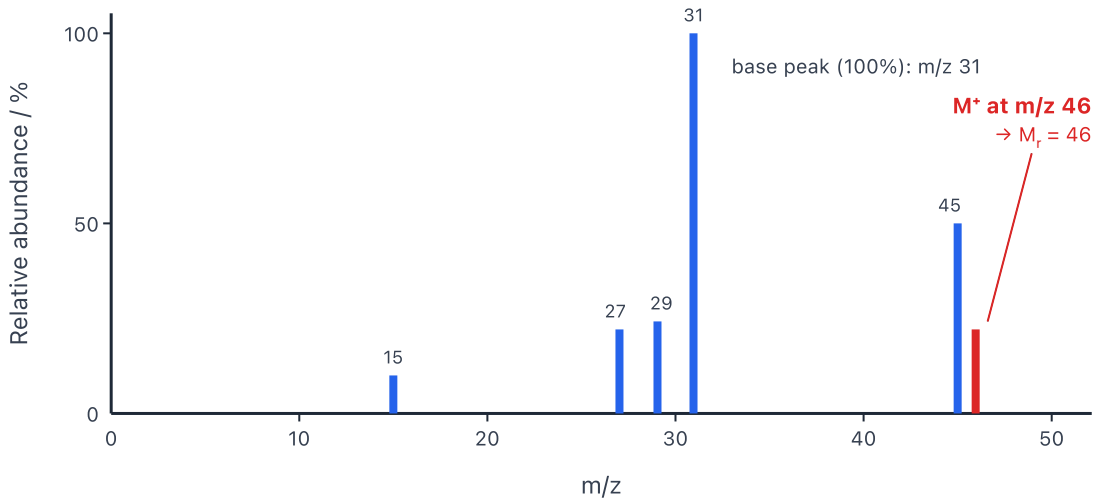


Every line of lithium and sodium appears in the unknown; potassium's lines do not → it contains Li^+ and Na^+ only.

Mass spectra. A mass spectrometer turns molecules into positive ions and sorts them by their mass-to-charge ratio (m/z). Features of the chart:

- **x-axis:** m/z (for ions with a 1+ charge this is the mass of the ion). **y-axis:** relative abundance (%).
- The **molecular ion peak (M^+)** is the peak with the **highest m/z** (furthest right). Its m/z value equals the **relative molecular mass, M_r**, of the compound.
- Peaks at lower m/z are **fragments** formed when the molecule breaks up. The pattern of fragments is a "fingerprint" which can be matched to a reference database.
- The tallest peak (the **base peak**, 100%) is the most common fragment — it is *not* necessarily the molecular ion.

Mass spectrum of ethanol, C_2H_5OH (M_r = 46)



Worked Example: An unknown liquid gives a mass spectrum whose highest-m/z peak is at 58. Reference data: propanone M_r 58, butane M_r 58, ethanol M_r 46, propan-1-ol M_r 60. What can you conclude?

The molecular ion gives M_r = 58, which rules out ethanol and propan-1-ol. Propanone and butane both have M_r 58, so the **fragmentation pattern** must be compared with the reference spectra to decide: the compound is whichever reference spectrum matches peak-for-peak.

Practicals linked to C4 (PAGs)

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PAG	Title (from the spec)	C4 activity
PAG C1	Reactivity trend	Displacement reactions of halogens with halides; reaction of metals with water and dilute HCl; metal/metal salt displacement; investigation of transition metals
PAG C5	Identification of species	Tests for cations (NaOH) and anions (silver nitrate, barium chloride); flame tests; testing unknown solutions; investigation of transition metals
PAG C7	Production of salts	Reactions of metals with acids and metal salts (the solutions formed are salts)
PAG C8	Measuring rates of reaction	Comparing rates/temperature changes for metals with acid and displacement reactions; transition metal catalysts

Quick Summary — Key Trends and Tests for C4

Topic	Key fact
Group 1 down the group	Reactivity ↑, melting point ↓ (outer electron lost more easily)
Group 7 down the group	Reactivity ↓, melting/boiling point ↑, colour darker (electron gained less easily)
Group 0	Full outer shell → unreactive, monatomic; boiling point ↑ down the group
Transition metals	High mp and density, less reactive, coloured ions with different charges, catalysts
Metal reactivity	K > Na > Li > Ca > Mg > Zn > Fe > Cu > Ag > Au
Gas tests	O_2 relights glowing splint · H_2 squeaky pop · CO_2 limewater milky · Cl_2 bleaches damp litmus

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NaOH tests	Ca^{2+} white · Zn^{2+} white, dissolves in excess · Cu^{2+} blue · Fe^{2+} green · Fe^{3+} orange-brown
Anion tests	CO_3^{2-} fizz + limewater · $Cl^-/Br^-/I^-$ white/cream/yellow with HNO_3 + $AgNO_3$ · SO_4^{2-} white with HCl + $BaCl_2$
Flame tests	Li^+ crimson · Na^+ yellow · K^+ lilac · Ca^{2+} orange-red · Cu^{2+} green
Mass spectrum	Molecular ion peak (highest m/z) = M_r

Specification checklist

Code	Learning outcome (summary)	Tier
C4.1a	Recall the simple physical and chemical properties of Groups 1, 7 and 0	F + H
C4.1b	Explain how properties of Groups 1, 7 and 0 depend on outer-shell electrons; predict properties from trends	F + H
C4.1c	General properties of transition metals and their compounds (mp, density, reactivity, coloured ions with different charges, catalysts)	F + H
C4.1d	Predict possible reactions and probable reactivity from position in the Periodic Table	F + H
C4.1e	Reactivity of metals with water or dilute acids related to tendency to form positive ions	F + H
C4.1f	Deduce an order of reactivity of metals from experimental results	F + H
C4.2a	Tests for oxygen, hydrogen, carbon dioxide and chlorine	F + H
C4.2b	Tests for aqueous cations (Ca^{2+}, Cu^{2+}, Fe^{2+}, Fe^{3+}, Zn^{2+} with NaOH) and anions (CO_3^{2-}, Cl^-, Br^-, I^-, SO_4^{2-})	F + H

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C4.2c	Describe how to perform a flame test	F + H
C4.2d	Identify species from test results	F + H
C4.2e	Interpret flame tests for Li^+, Na^+, K^+, Ca^{2+} and Cu^{2+}	F + H
C4.2f	Advantages of instrumental methods (sensitivity, accuracy, speed)	F + H
C4.2g	Interpret an instrumental result against reference data; features of a mass spectrum	F + H

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